

USUGLI, Russian Federation

WMO: 307640

Lat: 52.654N

Lon: 115.259E

Elev: 2119

StdP: 13.60

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.1	-38.8	-47.0	0.4	-42.5	-42.7	0.5	-38.3	15.3	6.0	13.3	3.5	0.0	180	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.9	84.6	65.5	81.1	64.3	77.9	63.0	68.7	79.3	66.8	76.6	65.0	74.2	5.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	101.5	73.2	63.5	94.8	70.7	61.6	88.5	69.2	34.4	79.7	32.7	76.9	31.3	74.6	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.1	11.9	-47.3	90.7	5.0	3.8	-50.9	93.5	-53.8	95.7	-56.7	97.8	-60.3	100.5		
(5)			-46.6	73.0	4.9	3.0	-50.2	75.1	-53.1	76.9	-55.8	78.5	-59.4	80.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	-17.0	-6.5	11.7	32.4	46.5	59.0	64.3	59.7	46.5	29.1	4.0	-14.3		
	DBStd	30.00	10.01	10.14	12.10	8.19	6.95	6.32	4.59	5.87	7.27	9.11	12.99	11.32		
	HDD50	9710	2077	1583	1187	529	153	8	0	6	148	646	1380	1993		
	HDD65	14146	2542	2003	1652	978	575	196	67	180	554	1111	1830	2458		
	CDD50	1115	0	0	0	1	43	277	443	306	45	0	0	0		
	CDD65	74	0	0	0	0	0	15	44	15	0	0	0	0		
	CDH74	1688	0	0	0	1	50	558	766	287	26	0	0	0		
	CDH80	415	0	0	0	0	4	153	199	57	2	0	0	0		
	CDH80	415	0	0	0	0	4	153	199	57	2	0	0	0		
(14) Wind	WSAvg	3.3	1.4	1.8	3.7	5.7	5.6	4.0	3.3	3.1	3.8	3.5	2.2	1.1		
(15) Precipitation	PrecAvg	14.1	0.1	0.1	0.1	0.4	1.1	2.8	3.5	3.3	1.7	0.4	0.3	0.2		
	PrecMax	18.7	0.5	0.4	0.4	1.2	2.5	5.7	7.7	7.3	4.0	1.4	1.0	0.6		
	PrecMin	8.6	0.0	0.0	0.0	0.0	0.0	0.4	0.8	1.3	0.3	0.0	0.0	0.0		
	PrecStd	2.9	0.1	0.1	0.1	0.3	0.6	1.5	1.7	1.5	0.9	0.3	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	30.4	50.4	68.0	80.3	89.6	89.5	85.8	77.5	62.3	43.2	24.8		
		MCWB	14.6	24.5	37.5	48.7	58.3	66.1	68.3	67.1	57.8	47.8	35.1	21.3		
	2%	DB	10.6	23.9	43.8	60.7	74.6	84.2	85.0	80.6	71.7	54.9	35.1	16.7		
		MCWB	8.6	19.8	33.8	44.8	54.7	63.5	66.8	65.0	56.2	42.8	29.1	14.5		
	5%	DB	5.0	17.9	38.0	55.1	69.9	80.2	81.6	76.9	66.9	49.6	29.4	9.2		
		MCWB	3.4	15.0	30.1	41.4	52.2	62.3	66.0	63.5	53.8	39.6	25.3	7.6		
	10%	DB	-0.1	13.0	32.7	50.3	65.0	76.5	78.2	73.2	62.4	44.9	24.3	3.0		
		MCWB	-1.2	10.8	26.6	38.6	49.6	60.8	64.9	62.3	51.6	36.4	21.2	2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	24.8	38.4	49.5	59.2	70.2	72.3	69.8	62.0	48.3	34.9	21.2		
		MCDB	18.0	29.9	49.7	65.7	76.8	84.0	83.1	78.7	71.1	61.3	43.0	24.3		
	2%	WB	8.5	19.7	34.1	45.4	55.8	66.4	69.5	67.2	58.8	43.5	29.6	14.4		
		MCDB	10.6	23.7	43.5	59.4	71.9	78.8	80.2	76.1	66.3	53.9	34.8	16.8		
	5%	WB	3.5	15.2	30.4	41.9	53.1	64.1	67.7	65.3	56.1	40.1	25.5	7.6		
		MCDB	5.0	18.0	37.7	54.2	67.5	75.4	77.8	73.7	64.6	48.8	29.4	9.1		
	10%	WB	-1.1	10.7	26.8	39.0	50.7	62.0	66.1	63.5	52.9	37.0	21.2	2.0		
		MCDB	0.0	12.9	32.6	49.9	63.8	73.6	75.4	71.2	61.5	44.0	24.2	2.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.7	34.6	33.1	27.6	29.0	29.2	25.9	24.7	27.5	25.9	25.9	23.4		
		MCDBR	29.0	36.2	33.7	37.7	40.3	38.2	34.2	32.3	37.1	33.4	29.0	26.6		
		MCWBR	26.8	32.3	24.4	23.1	21.7	19.2	17.3	17.3	21.6	21.9	23.1	24.4		
		MCDBR	28.7	35.5	33.0	36.2	36.5	31.8	28.3	26.5	31.0	31.6	28.5	26.6		
	5% WB	MCWBR	26.7	32.0	24.2	22.7	20.5	17.2	15.5	14.9	18.6	21.2	22.9	24.3		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.276	0.315	0.412	0.517	0.479	0.429	0.390	0.340	0.300	0.271	0.251		
	taud		2.232	2.201	2.191	2.052	1.816	2.038	2.248	2.344	2.427	2.442	2.304	2.185		
	Ebn at Noon		239	265	275	258	234	247	259	264	266	256	227	211		
	Edh at Noon		20	28	36	47	63	51	41	35	29	23	19	17		
(44) All-Sky Solar Radiation	RadAvg		433	796	1331	1659	1790	1898	1757	1504	1175	795	457	330		
	RadStd		11	16	30	102	118	144	157	88	95	51	23	10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page